Music4All: A New Music Database and its Applications

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Summary

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Music and society

- Music plays an important role in the development of the human being in a society.

- Children, as an example, can change their behavior, the way they think, and how they make relationships based on music.

Music information retrieval

- Music information retrieval (MIR) is gaining attention due to the widespread digital distribution of music.
- According to the IFPI Global Music Report 2019, there was an increase of $32,9\%$ in paid music streaming services from 2017 to 2018, with 255 million users.
- The consolidation of several streaming applications such as Spotify, Apple Music, and Pandora, emphasize the relevance of the area and the many opportunities in it.

Music information retrieval

- One of the goals of MIR is to present new methods that can efficiently and effectively retrieve and recommend songs from large databases of music content.

- In the literature, several databases have been presented, still missing some desirable requirement as:
  - A huge amount of music pieces.
  - Audio signal availability.
  - A great diversity of audio attributes.

Music4All

- Aiming at contributing with the MIR community, we present the Music4All database[1].
- This database contains information about songs, such as **metadata**, **tags**, **genre information**, **audio clips**, **lyrics**, etc. It also contains the **listening history** from users.
- Allowing researches on music recommendation, genre classification, auto-tagging, mood classification, etc.

[1] Send an e-mail to contact4music4all@gmail.com for more information.

Database - Music4All

- Music4all was designed to be a useful database in different domains of research.
- To achieve this, we collect information from several data sources.

Spotify YouTube last.fm

musixmatch Every Noise at Once

Database - Data acquisition

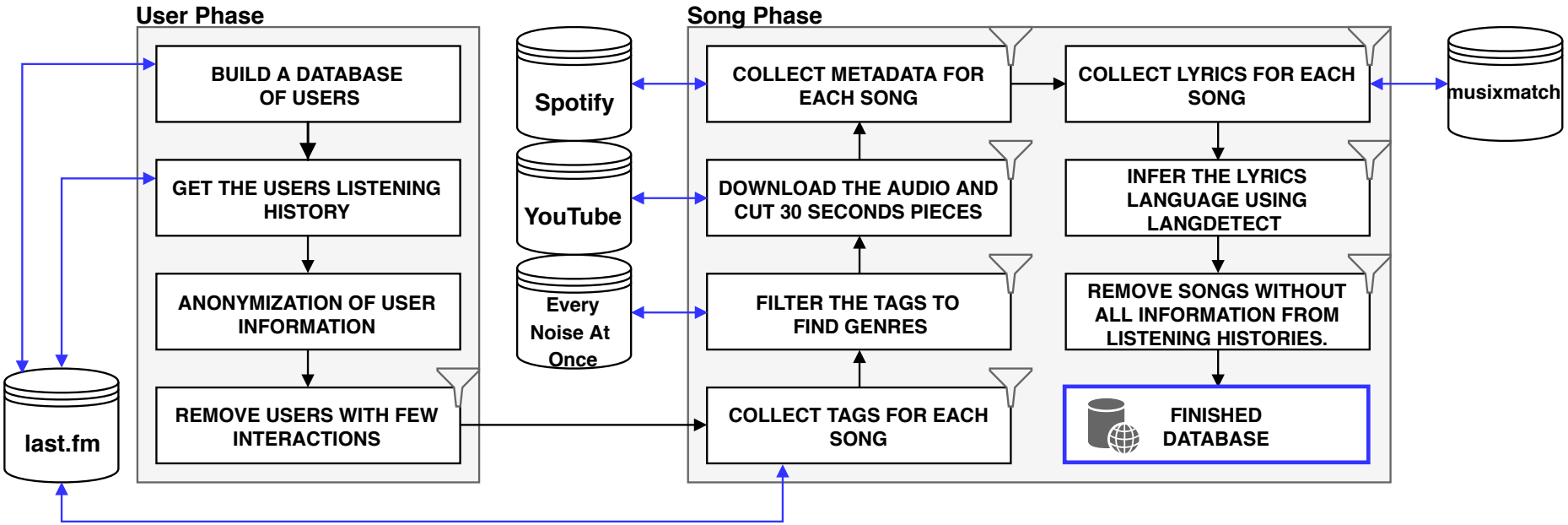


Figure: Summary of the development of the database.

Database - Data details

15,602 users

109,269 songs

16,269 artists

Songs in 46 languages

Database - Data details

30 seconds

Tags

Drums, Vocal, Live...

Genres

Rock, Metal...

Mood

Valency: 0.1
Enegy: 0.7

On song language

Applications

- To demonstrate how useful our database can be for the MIR community, we developed some applications that used our database.

- The applications are based on traditional MIR research fields: **music recommendation**, **genre classification**, and **mood classification**.

Applications - Music recommendation

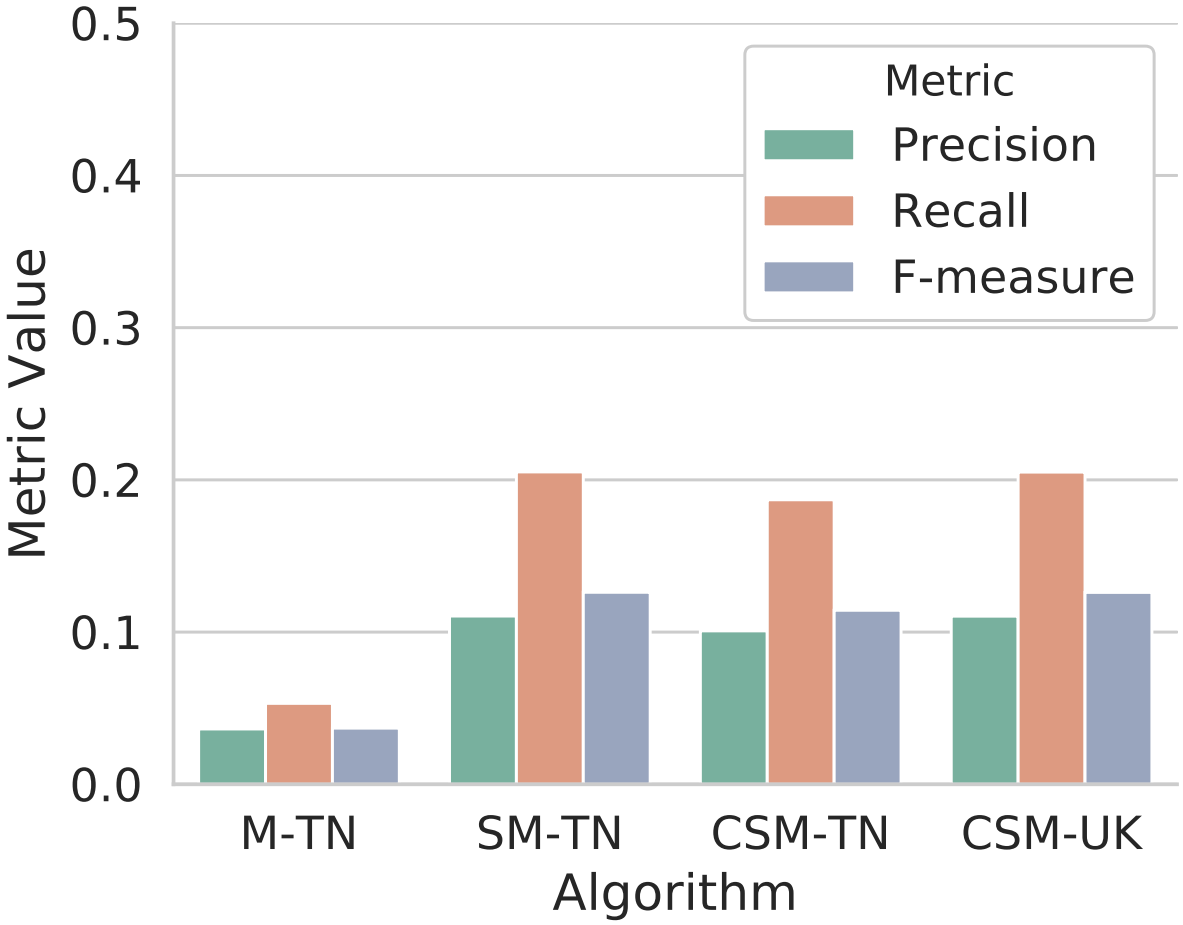
- The main goal of a **Music Recommender System** is to propose to a user interesting songs to listen to, based on the personal preference from the user.

- One of the applications in which our database can be applied, in this research area, is suggesting playlists to the user.

Applications - Music recommendation

- The listening history for each user was rearranged into sessions. Songs that were listened by the same user 60 minutes apart from each other were placed in different sessions.
- To represent the songs and capture the intrinsic features of the sequence in which they were listened by a given user we used embeddings captured by Music2Vec, an adaptation of Word2Vec for Music.
- We used 4 session-based recommender systems: Music2vec-TopN (M-TN), SessionMusic2vec-TopN (SM-TN), Context-Session-Music2vec-TopN(CSM-TN) and Context-Session-Music2vec-UserKNN (CSM-UK).

Applications - Music recommendation



Applications - Genre classification

- Automatic music genre classification is an important task in MIR.
- When performing music genre classification, audio-based approaches receives a significant amount of attention from the MIR community.
- Most of the work in the music genre classification area approaches a multi-class scenario, but in the real-world, music usually has more than one genre.

Applications - Genre classification

- We presented an architecture using Inception_v3 to perform genre classification using spectrograms.
- 5-fold cross-validation with artist filtering, resulting in 65 genres tags and 103,568 songs.
- Obtaining 0.2786 for the Accuracy and 0.3703 for the F-measure.

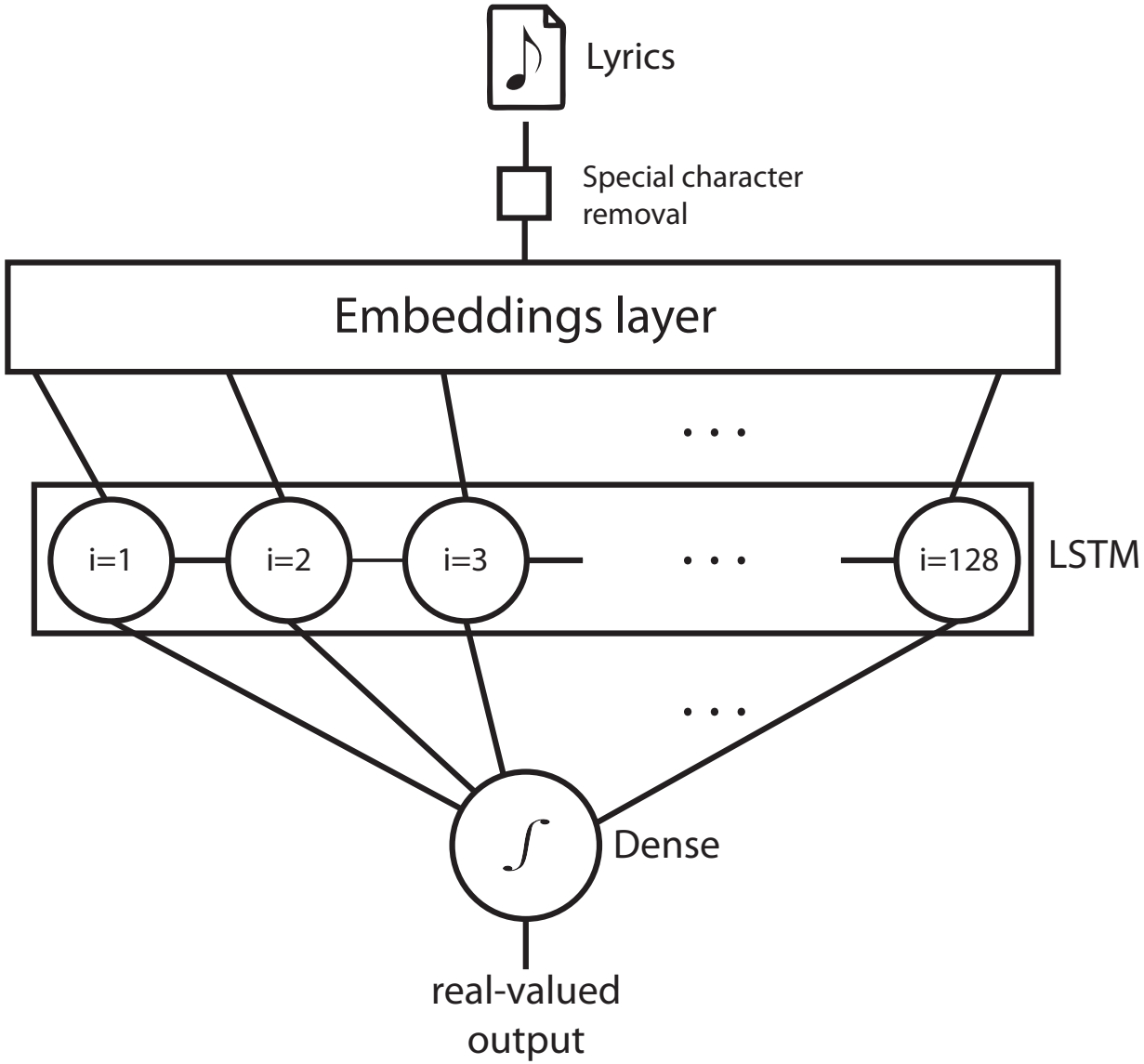
Applications - Mood classification

- Music Emotion Recognition (MER) has received increasing attention from the scientific community, as emotion is a fundamental attribute from a song.
- The most common way to approach this application is to categorize songs using specific emotions labels.
- Other way to approach the problem is by representing emotion, or aspects related to it, as continuous values.

Applications - Mood classification

- We performed mood classification inferring the value of valency and the energy (arousal).
- Only the lyrics of the songs were used as a source of information.
- To perform this task we use two Long Short-Term Memory (LSTM), one for each.

Applications - Mood classification



Applications - Mood classification

- Achieving in both cases a Mean Absolute Error close to 0.20 and the Pearson and Spearman correlation coefficients close to 0.34.

Comparison with others databases

Table: Some of the most important music databases.

Database	Number of Instances	Number of Classes	Available Content	Authoriz. Requested	Songs Lenght
RWC	100	10	Songs	Yes	Full
GTZAN	1,000	10	Songs	No	Songs
USPOP 2002	8,752	10	Songs	Yes	Full
ISMIR 2004	1,458	6	Songs	No	Full
Homburg	1,886	9	Songs	No	10 sec
Ballroom	698	8	Songs	No	30 sec
CAL500	500	174	Songs	No	Full
Holzapfel	N/A	6	Songs	Yes	Full
1517-artists	3,180	19	Songs	Yes	Full
LMD	3,227	10	Features	No	Full
Magnatagatune	21,642	188	Songs	No	30 sec
MSD	1,000,000	Variable	Features	No	Full
Music4All	109,269	Variable	Many	Yes	30 sec

Conclusion

- We presented Music4All, a new music database that can be used in several research areas related to MIR.
- This database contains different types of information about songs.
- We also presented some examples of applications that can be developed using Music4All.

Future Work

- Music4All was designed to be a database useful in distinct domains of research, and as future work we intent to explore it in many different applications.

- Future work will also involve research focused on multimodal applications.